REVISION WORKSHEET

UNITED ARAB EMIRATES
MINISTRY OF EDUCATION

الإمارات العربية المتحدة
وزارة التربية والتعليم

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Cycle 3 (10048) | Dibba Al Hisn | Sharjah Eastern

Practice Questions and Worked Answers

Grade 9 Advanced | Module 1 Lesson 3 | Measurement

All 30 questions are followed by answers. Study the reasoning, formula, substitution and units. Equivalent scientifically correct wording is acceptable.

Useful equations and guidance: Percentage uncertainty = (absolute uncertainty/measured value) × 100; mean = sum of readings/number of readings.

Term 1 2026–2027 | Textbook alignment: uploaded Grade 9 book, printed pages 13–16.

P01. Accuracy describes:

A. Closeness of repeated readings only B. The number of trials
C. The size of the object D. Closeness to an accepted reference value

Answer

D. Closeness to an accepted reference value.

P02. Precision describes:

A. Closeness of repeated readings to each other B. Closeness to the true value only
C. The absence of units D. The largest possible error

Answer

A. Closeness of repeated readings to each other.

P03. Which change mainly reduces random uncertainty?

A. Ignoring unusual results without investigation B. Repeating readings and averaging
C. Changing units only D. Using an uncalibrated instrument

Answer

B. Repeating readings and averaging.

P04. A balance reads 0.20 g with nothing on it. This suggests:

A. Perfect accuracy B. An error in SI definitions
C. A zero error D. An independent variable

Answer

C. A zero error.

P05. Which action helps avoid parallax error?

A. Looking from the side B. Standing farther away at an angle
C. Rounding every reading to one digit D. Reading the scale at eye level perpendicular to it

Answer

D. Reading the scale at eye level perpendicular to it.

P06. A length is reported as (40 ± 1) cm. Calculate its percentage uncertainty.

A. 5 % B. 1.25 %
C. 2.5 % D. -2.5 %

Answer

C. 2.5 %. Percentage uncertainty = (absolute uncertainty/measured value) × 100 = (1/40) × 100 = 2.5%.

P07. A scale has smallest divisions of 0.4 cm. Estimate the reading uncertainty as half a division.

A. 0.4 cm B. 0.1 cm
C. -0.2 cm D. 0.2 cm

Answer

D. 0.2 cm. Estimated uncertainty = smallest division/2 = 0.4/2 = ±0.2 cm.

P08. A standard length is 20 cm but an instrument reads 20.4 cm. Calculate the absolute measurement error.

A. 0.4 cm B. 0.8 cm
C. 0.2 cm D. -0.4 cm

Answer

A. 0.4 cm. Absolute error = |measured − reference| = |20.4 − 20| = 0.4 cm.

P09. Three repeated measurements are 12, 14 and 16 s. Calculate their mean.

A. 28 s B. 14 s
C. 7 s D. -14 s

Answer

B. 14 s. Mean = (12 + 14 + 16)/3 = 14 s. Averaging can reduce random variation.

P10. Repeated readings are 22, 24 and 26 cm. Calculate the range of the readings.

A. 8 cm B. 2 cm
C. 4 cm D. -4 cm

Answer

C. 4 cm. Range = largest − smallest = 26 − 22 = 4 cm.

P11. Define a measurement.

Answer

A measurement compares an unknown quantity with an agreed standard and reports a value with a unit.

P12. Why should uncertainty be reported with a measurement?

Answer

It indicates the range or limitation supported by the instrument and method; measurements are not perfectly exact.

P13. Distinguish random error from systematic error.

Answer

Random error causes unpredictable scatter; systematic error shifts results in a consistent way, such as a zero offset.

P14. Explain why a precise instrument can still give inaccurate results.

Answer

Poor calibration or a systematic offset can make readings tightly grouped but far from the accepted value.

P15. What is instrument resolution?

Answer

The smallest change in a quantity that the instrument can distinguish or display.

P16. A length is reported as (60 ± 1.5) cm. Calculate its percentage uncertainty.

Answer

Percentage uncertainty = (absolute uncertainty/measured value) × 100 = (1.5/60) × 100 = 2.5%.

P17. A length is reported as (80 ± 2) cm. Calculate its percentage uncertainty.

Answer

Percentage uncertainty = (absolute uncertainty/measured value) × 100 = (2/80) × 100 = 2.5%.

P18. A scale has smallest divisions of 0.6 cm. Estimate the reading uncertainty as half a division.

Answer

Estimated uncertainty = smallest division/2 = 0.6/2 = ±0.3 cm.

P19. A scale has smallest divisions of 0.8 cm. Estimate the reading uncertainty as half a division.

Answer

Estimated uncertainty = smallest division/2 = 0.8/2 = ±0.4 cm.

P20. A standard length is 30 cm but an instrument reads 30.6 cm. Calculate the absolute measurement error.

Answer

Absolute error = |measured − reference| = |30.6 − 30| = 0.6 cm.

P21. A standard length is 40 cm but an instrument reads 40.8 cm. Calculate the absolute measurement error.

Answer

Absolute error = |measured − reference| = |40.8 − 40| = 0.8 cm.

P22. Three repeated measurements are 13, 15 and 17 s. Calculate their mean.

Answer

Mean = (13 + 15 + 17)/3 = 15 s. Averaging can reduce random variation.

P23. Three repeated measurements are 14, 16 and 18 s. Calculate their mean.

Answer

Mean = (14 + 16 + 18)/3 = 16 s. Averaging can reduce random variation.

P24. Repeated readings are 23, 25 and 27 cm. Calculate the range of the readings.

Answer

Range = largest − smallest = 27 − 23 = 4 cm.

P25. Repeated readings are 24, 26 and 28 cm. Calculate the range of the readings.

Answer

Range = largest − smallest = 28 − 24 = 4 cm.

P26. A length is (10.0 ± 0.2) cm. Does a second value of 10.1 cm agree within this uncertainty? Explain.

Answer

Yes. The stated interval is 9.8–10.2 cm, which includes 10.1 cm.

P27. A balance consistently reads 0.50 g too high. How should you improve the measurements?

Answer

Check and correct its zero/calibration. Repeating and averaging alone will not remove the systematic offset.

P28. Describe how to measure the thickness of one sheet of paper more reliably with a ruler.

Answer

Measure a stack of many sheets, then divide the stack thickness by the number of sheets. This reduces relative uncertainty from the ruler resolution.

P29. Readings are 5.1, 5.1 and 5.2 cm when the accepted value is 6.0 cm. Discuss accuracy and precision.

Answer

The readings are relatively precise because they are close together, but inaccurate because they are far from 6.0 cm.

P30. A student removes a reading only because it disagrees with the hypothesis. Evaluate the decision.

Answer

It is not justified. Investigate a possible procedural or instrument error, keep a transparent record and exclude data only for a documented valid reason.